

Introduction to mathematics for economists

Exercise Sheet #2 — Linear Algebra

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During the tutorials, we will solve exercises 1, 3, 5, 8, 10, and 12. So you must prioritize these. You will be provided a full answer sheet.

Vector Spaces, Span, and Independence

Subspace

A subset $W \subseteq \mathbb{R}^n$ is a **linear subspace** if it is nonempty and closed under the two linear operations:

$$\mathbf{u}, \mathbf{v} \in W \implies \mathbf{u} + \mathbf{v} \in W, \quad \mathbf{u} \in W, c \in \mathbb{R} \implies c\mathbf{u} \in W.$$

Equivalently: W contains $\mathbf{0}$ and every linear combination of its elements.

Exercise 1 : Subspaces, cones, and why “linear” earns its name

- For each set below, decide whether it is a subspace of their definition space. Prove or disprove by giving a counter-example. If it is a subspace, give a basis and its dimension.
 - $W_1 = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$.
 - $W_2 = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$.
 - $W_3 = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$ (the non-negative quadrant).
 - $W_4 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.
- The set of feasible consumption bundles is the non-negative quadrant $\mathbb{R}_+^n = \{x : x_i \geq 0 \forall i\}$. Explain in one or two sentences why this is a **cone** (i.e. only multiplications by positive scalars allowed) but not a vector space, and why a budget line $\{x : p \cdot x = m\}$ (with $m > 0$) is not one either.
- Let U, W be subspaces of $\mathbb{R}^n$. Show that $U \cap W$ is always a subspace, but that $U \cup W$ generally is **not**. Give a concrete counterexample for the union in $\mathbb{R}^2$.

i Solution

Question 1.

(a) W_1 : *subspace*. It contains $\mathbf{0} = (0, 0)$ since $0 + 2 \cdot 0 = 0$. If (x_1, y_1) and (x_2, y_2) both satisfy the constraint, then $(x_1 + x_2) + 2(y_1 + y_2) = (x_1 + 2y_1) + (x_2 + 2y_2) = 0$, so the sum stays in W_1 ; and $c(x, y)$ gives $cx + 2cy = c(x + 2y) = 0$. Both closures hold. Geometrically $W_1 = \text{span}\{(-2, 1)\}$, a line through the origin, $\dim W_1 = 1$.

(b) W_2 : *not a subspace*. Every subspace must contain $\mathbf{0}$, but $0 + 2 \cdot 0 = 0 \neq 3$, so $\mathbf{0} \notin W_2$. (Closure fails too: $(3, 0), (1, 1) \in W_2$ but their sum $(4, 1)$ gives $4 + 2 = 6 \neq 3$.) W_2 is an *affine* line — a translate of W_1 — not a linear one.

(c) W_3 : *not a subspace*. It is closed under addition (two non-negative vectors add to a non-negative vector) but **not** under scalar multiplication: take $v = (1, 1) \in W_3$ and $c = -1$, then $cv = (-1, -1) \notin W_3$. One broken axiom sinks it. W_3 is a *cone* — closed under non-negative scaling only.

(d) W_4 : *subspace*. Contains $\mathbf{0}$; closures follow exactly as in (a) with the constraint $x + y + z = 0$. It is the kernel of the map $(x, y, z) \mapsto x + y + z$, a plane through the origin. A basis is $\{(1, -1, 0), (1, 0, -1)\}$ (both satisfy the constraint and are not multiples of each other), so $\dim W_4 = 2$. ✓

Question 2.

You can add two consumption bundles and scale a bundle up, and you stay feasible — but you cannot consume *negative* quantities, so multiplying a bundle by a negative scalar leaves the set. Closed under non-negative scaling but not all scaling: that is the definition of a cone, not a vector space. The budget line $\{x : p \cdot x = m\}$ fails even earlier: since $p \cdot \mathbf{0} = 0 \neq m$, it does not contain the origin, so it is affine, not linear.

Question 3.

Intersection. $\mathbf{0} \in U$ and $\mathbf{0} \in W$, so $\mathbf{0} \in U \cap W$ (nonempty). If $x, y \in U \cap W$, then $x, y \in U \Rightarrow x + y \in U$, and $x, y \in W \Rightarrow x + y \in W$, hence $x + y \in U \cap W$; scalar multiples are handled identically. So $U \cap W$ is a subspace.

Union. Take $U = \{(t, 0) : t \in \mathbb{R}\}$ (the x -axis) and $W = \{(0, s) : s \in \mathbb{R}\}$ (the y -axis). Then $(1, 0) \in U \subset U \cup W$ and $(0, 1) \in W \subset U \cup W$, but $(1, 0) + (0, 1) = (1, 1) \notin U \cup W$. Closure under addition fails, so the union is not a subspace. (Intuitively: a union of two lines is not flat.)

Exercise 2 : Spanning, independence, and complete markets

Consider a one-period economy with two future **states of the world**, s_1 and s_2 . An asset is described by the vector of its payoffs across states: an asset paying θ_1 in state s_1 and θ_2 in state s_2 is the vector $(\theta_1, \theta_2) \in \mathbb{R}^2$. A **portfolio** is a linear combination of assets (holding c_j units of asset j), and its payoff vector is the same linear combination of payoff vectors. A market is called **complete** if every conceivable payoff profile (θ_1, θ_2) can be manufactured by some portfolio.

Take the risk-free bond $a_1 = (1, 1)$ (pays 1 in each state) and the Arrow security $a_2 = (2, 0)$ (pays 2 in state s_1 , nothing in state s_2).

1. Are a_1 and a_2 linearly independent? Do they span $\mathbb{R}^2$? Conclude whether this two-asset market is complete.
2. Now add a third asset $a_3 = (0, 3)$. Explain, without computing, why $\{a_1, a_2, a_3\}$ must be linearly dependent. Then find an explicit non-trivial dependence relation $\alpha a_1 + \beta a_2 + \gamma a_3 = 0$.
3. Find the portfolio in a_1 and a_2 that **replicates** a_3 . In one sentence, relate this to the idea that a redundant asset carries no new insurance possibilities.
4. Give a basis for $\mathbb{R}^2$ drawn from these assets and state $\dim \mathbb{R}^2$. State in one line what “complete market” means in the vocabulary of span and dimension.

i Solution**Question 1.**

Suppose $c_1 a_1 + c_2 a_2 = (0, 0)$, i.e. $c_1(1, 1) + c_2(2, 0) = (c_1 + 2c_2, c_1) = (0, 0)$. The second coordinate forces $c_1 = 0$, and then the first forces $c_2 = 0$. Only the trivial combination vanishes, so a_1, a_2 are **linearly independent**. Two independent vectors in $\mathbb{R}^2$ automatically span the whole space (a linearly independent set of size $\dim V$ is a basis), so any payoff profile is attainable: the market is **complete**. ✓

Question 2.

Three vectors in a two-dimensional space can never be independent — any set of more than $\dim V$ vectors is dependent. To find the relation, write a_3 in terms of a_1, a_2 : solve $\alpha(1, 1) + \beta(2, 0) = (0, 3)$. The second coordinate gives $\alpha = 3$; the first gives $3 + 2\beta = 0$, so $\beta = -\frac{3}{2}$. Hence

$$3a_1 - \frac{3}{2}a_2 - a_3 = 0.$$

(Check: $3(1, 1) - \frac{3}{2}(2, 0) - (0, 3) = (3, 3) - (3, 0) - (0, 3) = (0, 0)$. ✓)

Question 3.

Rearranging the relation, $a_3 = 3a_1 - \frac{3}{2}a_2$: hold 3 units of the risk-free bond and **short** $\frac{3}{2}$ units of the Arrow security. The payoff of this portfolio equals a_3 's in every state, so a_3 opens up no state-contingent transfer that a_1 and a_2 could not already produce — it is redundant.

Question 4.

$\{a_1, a_2\}$ is a basis (independent and spanning), and $\dim \mathbb{R}^2 = 2$. **Completeness in linear-algebra terms:** the asset payoff vectors span the entire state space $\mathbb{R}^S$ (= can achieve any desired payoff profile across future states); equivalently, the payoff matrix

$$P = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

has rank equal to the number of states.

Orthogonality and Projections

Dot Product, Orthogonality, and Projection

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the **dot product** is $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$, and $\mathbf{u}, \mathbf{v}$ are **orthogonal** iff $\mathbf{u} \cdot \mathbf{v} = 0$.

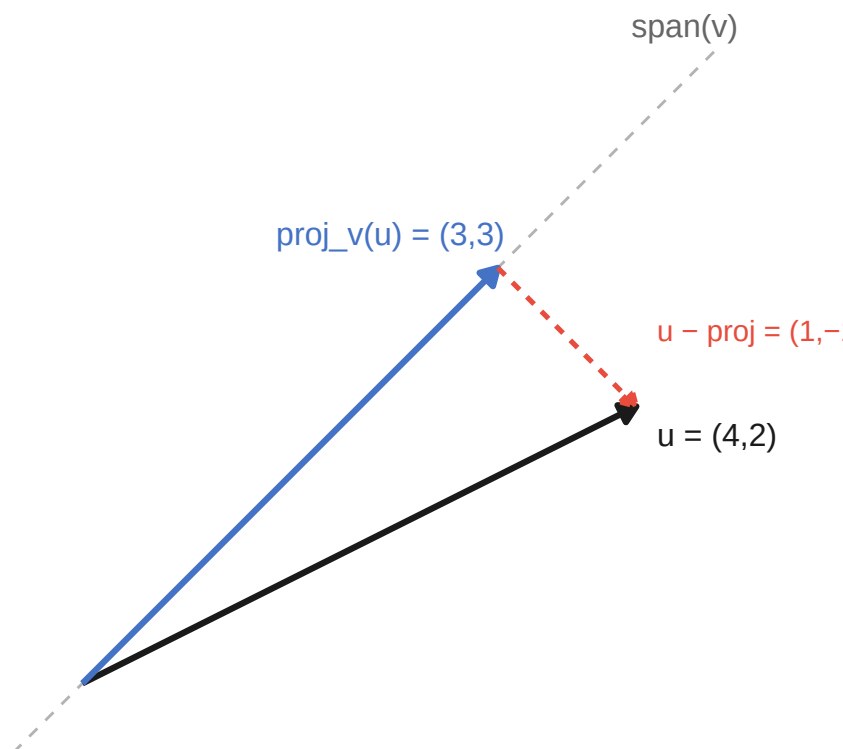
The **orthogonal projection** of u onto the line spanned by $v \neq 0$ is

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v.$$

More generally, for a subspace $S = \text{col}(A)$ the projection of u onto S is the unique $\hat{u} \in S$ whose residual is orthogonal to all of S ; solving $A^\top(u - Ac) = 0$ gives $\hat{u} = A(A^\top A)^{-1}A^\top u = P_S u$, with $P_S = A(A^\top A)^{-1}A^\top$.

Exercise 3 : From a line to a subspace to the projection matrix

1. Compute $\text{proj}_v(u)$ for $u = (4, 2)$ and $v = (1, 1)$. Verify explicitly that the residual $u - \text{proj}_v(u)$ is orthogonal to v .
2. Derive the projection formula from scratch. Try not to check the lecture notes: this is just to remind you that the formula is not really worth remembering because it's derived easily :)
3. Now move up to a plane in $\mathbb{R}^3$. Project $u = (1, 2, 0)$ onto $S = \text{span}\{a_1, a_2\}$ with $a_1 = (1, 0, 1)$, $a_2 = (0, 1, 1)$. Use $A = [a_1 \ a_2]$, solve the normal equations $A^\top A c = A^\top u$, and report $\hat{u}$ and the residual. Check the residual is orthogonal to **both** a_1 and a_2 . *Hint: you will need to invert a certain matrix.*
4. Form the projection matrix $P_S = A(A^\top A)^{-1}A^\top$ for the previous part. Verify that (i) P_S is symmetric, (ii) $P_S u$ reproduces the $\hat{u}$ you found, and (iii) $\text{tr}(P_S) = \dim S = 2$.



Solution

Question 1.

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v = \frac{4+2}{1+1}(1, 1) = 3(1, 1) = (3, 3).$$

Residual: $u - \text{proj}_v(u) = (4, 2) - (3, 3) = (1, -1)$, and $(1, -1) \cdot (1, 1) = 1 - 1 = 0$. Orthogonal, as required. ✓

Question 2.

The projection lies on the line, so it equals cv for some scalar c . The defining property is that the leftover $u - cv$ points perpendicular to v :

$$(u - cv) \cdot v = 0 \implies u \cdot v - c(v \cdot v) = 0 \implies c = \frac{u \cdot v}{v \cdot v}.$$

So $\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v$. The formula is not a thing to remember; it is the answer to “which multiple of v leaves an orthogonal residual?”

Question 3.

With $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $u = (1, 2, 0)$:

$$A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad A^T u = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 2 + 1 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 2 + 1 \cdot 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Solve $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ using $(A^T A)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$:

$$c = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Hence $\hat{u} = 0 \cdot a_1 + 1 \cdot a_2 = (0, 1, 1)$ and the residual is $u - \hat{u} = (1, 2, 0) - (0, 1, 1) = (1, 1, -1)$. Orthogonality check: $(1, 1, -1) \cdot a_1 = 1 + 0 - 1 = 0$ and $(1, 1, -1) \cdot a_2 = 0 + 1 - 1 = 0$. Both vanish. ✓

Question 4.

$$A(A^T A)^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{pmatrix},$$

$$P_S = A(A^T A)^{-1} A^T = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

(i) Symmetric by inspection. (ii) $P_S u = \frac{1}{3}(2 - 2 + 0, -1 + 4 + 0, 1 + 2 + 0) = \frac{1}{3}(0, 3, 3) = (0, 1, 1) = \hat{u}$. ✓ (iii) $\text{tr}(P_S) = \frac{1}{3}(2 + 2 + 2) = 2 = \dim S$. ✓

Exercise 4 : Properties of projection matrices

Let $S = \text{col}(A) \subseteq \mathbb{R}^n$ with A of full column rank, and $P = A(A^T A)^{-1} A^T$.

1. Prove P is **idempotent**: $P^2 = P$. (Work directly from the formula.)
2. Prove P is **symmetric**: $P^T = P$.

3. Let $M = I - P$. Show M is also symmetric and idempotent, and that M projects onto the orthogonal complement $S^\perp$.
4. **The eigenvalues of P** , and why $\text{tr}(P) = \dim S$. We build this up in small steps.
- Suppose $Pv = \lambda v$ for some $v \neq 0$. Apply P once more to both sides, use $P^2 = P$ from part 1, and deduce that $\lambda^2 = \lambda$. Conclude that the **only** possible eigenvalues of P are 0 and 1.
 - Show that all vectors of S have eigenvalue 1, and that if a vector has eigenvalue 1 then it must belong to S . Show similarly that the vectors with eigenvalue 0 are precisely the vectors of $S^\perp$. (*Hint: use part 3, and the fact that P fixes every vector already in S and sends every vector of $S^\perp$ to 0.*)
 - You may use the following fact: **every symmetric matrix is diagonalisable** (this is the *spectral theorem*). Combining this with (a) and (b), explain why P has exactly $\dim S$ eigenvalues equal to 1 and $n - \dim S$ eigenvalues equal to 0.
 - Use (c) to conclude

$$\text{tr}(P) = \text{rank}(P) = \dim S.$$

i Solution

Question 1.

$$P^2 = A(A^\top A)^{-1} \underbrace{A^\top A(A^\top A)^{-1}}_{=I} A^\top = A(A^\top A)^{-1} A^\top = P.$$

The inner $A^\top A(A^\top A)^{-1}$ collapses to the identity. ✓

Question 2.

$$P^\top = (A(A^\top A)^{-1} A^\top)^\top = A((A^\top A)^{-1})^\top A^\top.$$

Since $A^\top A$ is symmetric, so is its inverse, hence $((A^\top A)^{-1})^\top = (A^\top A)^{-1}$ and $P^\top = P$. ✓

Question 3.

$M^\top = (I - P)^\top = I - P^\top = I - P = M$, so M is symmetric. And

$$M^2 = (I - P)^2 = I - 2P + P^2 = I - 2P + P = I - P = M,$$

using $P^2 = P$. For any u , $Mu = u - Pu$ is the residual, which is orthogonal to S (that is what the projection guarantees), so $Mu \in S^\perp$; and if $u \in S^\perp$ then $Pu = 0$, so $Mu = u$. A symmetric idempotent that fixes $S^\perp$ and kills S is exactly the projection onto $S^\perp$. ✓

Question 4.

(a) *Eigenvalues can only be 0 or 1.* Suppose $Pv = \lambda v$ with $v \neq 0$. Apply P to both sides:

$$P(Pv) = P(\lambda v) = \lambda Pv = \lambda^2 v.$$

But the left-hand side is $P^2 v = Pv = \lambda v$, using $P^2 = P$ from part 1. So $\lambda^2 v = \lambda v$; since $v \neq 0$, this forces $\lambda^2 = \lambda$, i.e. $\lambda(\lambda - 1) = 0$. Hence $\lambda = 0$ or $\lambda = 1$ — no other value is possible. ✓

(b) *Which vectors sit in each eigenvalue.* For $\lambda = 1$: the equation reads $Pv = v$, so v is a vector that P leaves **unchanged**. Every output of P lands in S (that is the image), so $Pv = v$ forces $v \in S$. Conversely, if $v \in S$ then projecting a vector already in S returns it untouched, $Pv = v$. So the set of eigenvectors for $\lambda = 1$ is exactly S — an eigenspace of dimension $\dim S$. For $\lambda = 0$: the equation reads $Pv = 0$. By part 3,

$Pv = 0 \iff Mv = v \iff v \in S^\perp$. So the eigenvectors for $\lambda = 0$ are exactly $S^\perp$ — an eigenspace of dimension $n - \dim S$.

(c) *Counting the eigenvalues.* By the spectral theorem, the symmetric matrix P is diagonalisable, so P has n independent linear vectors, which implies $\mathbb{R}^n$ is spanned entirely by eigenvectors of P . From (a) the only eigenvalues are 1 and 0, and from (b) their respective eigenspaces are S (dimension $\dim S$) and $S^\perp$ (dimension $n - \dim S$). Therefore, there must be $\dim S$ eigenvectors with eigenvalue 1 (otherwise, S is not spanned by these vectors) and similarly $n - \dim S$ eigenvalues equal 0.

(d) *The trace.* Summing the eigenvalues,

$$\text{tr}(P) = \sum_i \lambda_i = (\dim S) \cdot 1 + (n - \dim S) \cdot 0 = \dim S.$$

For a diagonalisable matrix the rank equals the number of **nonzero** eigenvalues, which here is the number of 1's, namely $\dim S$. So $\text{tr}(P) = \text{rank}(P) = \dim S$. ✓

Linear Maps, Rank, and the Rank-Nullity Theorem

Image, Kernel, Rank

For a linear map $f(x) = Ax$ with $A \in \mathbb{M}_{m \times n}$:

$$\text{Im}(f) = \text{span}\{c_1(A), \dots, c_n(A)\} \subseteq \mathbb{R}^m, \quad \ker(f) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n,$$

and $\text{Rank}(A) = \dim \text{Im}(f) = \text{number of linearly independent columns}$.

Rank–Nullity

For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$,

$$\underbrace{\text{Rank}(f)}_{\dim \text{Im}(f)} + \underbrace{\text{Null}(f)}_{\dim \ker(f)} = n = \dim(\text{domain}).$$

Every independent constraint the map imposes costs one degree of freedom; the nullity is whatever freedom survives.

Exercise 5 : Image, kernel, rank — and the degrees of freedom in OLS

Let

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 5 \end{pmatrix} : \mathbb{R}^3 \rightarrow \mathbb{R}^3.$$

1. Find a basis for $\text{Im}(A)$ and state $\text{Rank}(A)$. (*Hint: start by checking dependence.*)
2. Find a basis for $\ker(A)$ and state the nullity.
3. Verify the rank–nullity theorem for A .
4. Is $f(x) = Ax$ injective? Surjective onto $\mathbb{R}^3$? For a given b , how many solutions can $Ax = b$ have — 0, 1, or ∞ — and on what does the count depend?

i Solution

Question 1.

The columns are $c_1 = (1, 1, 2)$, $c_2 = (1, 2, 3)$, $c_3 = (2, 3, 5)$. Notice $c_3 = c_1 + c_2$, so column 3 is redundant. The remaining c_1, c_2 are not scalar multiples of each other, hence independent. A basis for the image is $\{(1, 1, 2), (1, 2, 3)\}$ and $\text{Rank}(A) = 2$.

Question 2.

Solve $Ax = 0$. Row-reduce: $R_2 \leftarrow R_2 - R_1$ gives $(0, 1, 1)$; $R_3 \leftarrow R_3 - 2R_1$ gives $(0, 1, 1)$; then $R_3 \leftarrow R_3 - R_2$ gives $(0, 0, 0)$. The surviving equations are

$$x_1 + x_2 + 2x_3 = 0, \quad x_2 + x_3 = 0.$$

From the second, $x_2 = -x_3$; substituting into the first, $x_1 - x_3 + 2x_3 = 0 \Rightarrow x_1 = -x_3$. So $x = x_3(-1, -1, 1)$, and

$$\ker(A) = \text{span}\{(-1, -1, 1)\}, \quad \dim \ker(A) = 1.$$

(Consistent with $c_1 + c_2 - c_3 = 0$, which says $A(1, 1, -1)^\top = 0$.)

Question 3.

$\text{Rank}(A) + \dim \ker(A) = 2 + 1 = 3 = n$. ✓

Question 4.

Not injective: $\ker(A) \neq \{0\}$, so distinct inputs (differing by a kernel vector) share an output. Not surjective onto $\mathbb{R}^3$: the image is a 2-dimensional plane inside $\mathbb{R}^3$. Consequently $Ax = b$ has **no** solution when $b \notin \text{Im}(A)$, and **infinitely many** (a line $x^* + \ker(A)$) when $b \in \text{Im}(A)$. It never has exactly one solution — that would require $\ker(A) = \{0\}$.

Exercise 6 : The Leontief input–output economy

An economy has two sectors, **Agriculture** (1) and **Manufacturing** (2). Producing one unit of agricultural output requires 0.1 units of agriculture and 0.3 units of manufacturing as inputs; producing one unit of manufacturing output requires 0.4 units of agriculture and 0.2 units of manufacturing. Households demand 50 units of agriculture and 60 of manufacturing as final goods. Writing a_{ij} for the units of good i needed to make one unit of good j ,

$$A = \begin{pmatrix} 0.1 & 0.4 \\ 0.3 & 0.2 \end{pmatrix}, \quad d = \begin{pmatrix} 50 \\ 60 \end{pmatrix}.$$

1. Write the material-balance equation $x = Ax + d$ and explain, in one or two sentences, why the presence of x on both sides means you cannot solve one sector before the other.
2. Solve for total output $x = (I - A)^{-1}d$. Compute $(I - A)^{-1}$ with the 2×2 inverse formula.
3. Verify that your x satisfies $x - Ax = d$ coordinate by coordinate.
4. The $(1, 1)$ entry of $(I - A)^{-1}$ is the **output multiplier** for agriculture. Read it off, and explain why a one-unit rise in final demand for agriculture raises total agricultural output by *more* than one unit.

i Solution

Question 1.

Total output of each good must cover both what other sectors consume as inputs, $\sum_j a_{ij}x_j$, and final demand d_i : $x_i = \sum_j a_{ij}x_j + d_i$, i.e. $x = Ax + d$. Because agriculture's required output depends on manufacturing's (manufacturing uses agricultural inputs) and vice versa,

there is no sector you can pin down first — it is a simultaneous system, a fixed point, not a recursive calculation.

Question 2.

$$I - A = \begin{pmatrix} 0.9 & -0.4 \\ -0.3 & 0.8 \end{pmatrix}, \quad \det(I - A) = (0.9)(0.8) - (-0.4)(-0.3) = 0.72 - 0.12 = 0.6.$$

$$(I - A)^{-1} = \frac{1}{0.6} \begin{pmatrix} 0.8 & 0.4 \\ 0.3 & 0.9 \end{pmatrix}.$$

$$x = (I - A)^{-1}d = \frac{1}{0.6} \begin{pmatrix} 0.8 & 0.4 \\ 0.3 & 0.9 \end{pmatrix} \begin{pmatrix} 50 \\ 60 \end{pmatrix} = \frac{1}{0.6} \begin{pmatrix} 40 + 24 \\ 15 + 54 \end{pmatrix} = \frac{1}{0.6} \begin{pmatrix} 64 \\ 69 \end{pmatrix} \approx \begin{pmatrix} 106.7 \\ 115.0 \end{pmatrix}.$$

Question 3.

$(Ax)_1 = 0.1(106.7) + 0.4(115.0) = 10.67 + 46.0 = 56.67$, so $x_1 - (Ax)_1 = 106.7 - 56.67 = 50 = d_1$. ✓

$(Ax)_2 = 0.3(106.7) + 0.2(115.0) = 32.0 + 23.0 = 55.0$, so $x_2 - (Ax)_2 = 115.0 - 55.0 = 60 = d_2$. ✓

Exactly 50 units of agriculture and 60 of manufacturing reach final consumers; the rest is absorbed internally as inputs.

Question 4.

The $(1, 1)$ entry is $0.8/0.6 \approx 1.33$: one extra unit of final demand for agriculture raises total agricultural output by about 1.33 units. It exceeds 1 because meeting the extra demand requires producing more agriculture, which consumes manufacturing, which itself consumes agriculture, and so on — the feedback loop adds to the direct effect.

Determinants, Invertibility, and Linear Systems

Determinant — Key Facts

The determinant $\det : \mathbb{M}_{n \times n} \rightarrow \mathbb{R}$ satisfies $\det(AB) = \det A \det B$, $\det(A^\top) = \det A$, $\det(cA) = c^n \det A$, and equals the product of the diagonal for triangular matrices. Geometrically $|\det A|$ is the volume-scaling factor of the map, and its sign encodes orientation. The central equivalence:

$$\det A \neq 0 \iff \text{Rank}(A) = n \iff A^{-1} \text{ exists} \iff Ax = b \text{ has a unique solution for every } b.$$

Exercise 7 : Computing and reading determinants

We know this exercise is painful (computing co-factors is boring for all of us), but it's good to have done it a few times before you begin using your computer :)

1. Compute $\det B$ for $B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ by cofactor expansion along the first row.
2. Suppose A is 3×3 with $\det A = 5$. Using only the properties above (no entries needed), compute $\det(2A)$, $\det(A^\top)$, $\det(A^{-1})$, and $\det(A^2)$.
3. For the map $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$, compute $\det A$, state the factor by which it scales area, and say whether it preserves or reverses orientation.
4. Prove that if two columns of a square matrix are equal, then its determinant is 0.

Solution

Question 1.

Expanding along row 1:

$$\det B = 2 \det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} - 1 \det \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} + 0 = 2(4 - 1) - 1(2 - 0) = 6 - 2 = 4.$$

Question 2.

$$\det(2A) = 2^3 \det A = 8 \cdot 5 = 40, \quad \det(A^\top) = \det A = 5,$$

$$\det(A^{-1}) = \frac{1}{\det A} = \frac{1}{5}, \quad \det(A^2) = (\det A)^2 = 25.$$

Question 3.

A is upper-triangular, so $\det A = 3 \cdot 2 = 6$. It scales areas by a factor of 6, and since $\det A > 0$ it **preserves** orientation (no flip).

Question 4.

Let A have two identical columns. Swapping those two columns leaves A literally unchanged, so $\det A$ is unchanged. But swapping any two columns flips the sign of the determinant (the alternating property), so $\det A = -\det A$, forcing $\det A = 0$. Equal columns are a special case of linearly dependent columns; more generally, dependent columns mean $\text{Rank}(A) < n$, which by the central equivalence is exactly $\det A = 0$, i.e. A is singular.

Exercise 8 : Linear Systems, Cramer's Rule, and IS–LM Equilibria

A stylised macroeconomy is summarised by two market-clearing conditions in the two unknowns: output Y and the interest rate i . The goods market (IS, representing the investment-saving tradeoff) and the money market (LM, representing the liquidity preference–money supply tradeoff) yield the following system:

$$\text{IS: } Y + \beta i = \alpha, \quad \text{LM: } kY - hi = \frac{M}{P}$$

where α (which rises with government spending G), $\beta, k, h > 0$, and $M/P > 0$ are structural parameters.

Part I: The Linear System

1. Write the system in matrix form $Az = b$ with $z = (Y, i)^\top$, b the outputs of the goods and money market and A the linear map that relates the goods z to the outputs.
2. Compute $\det A$. With the given sign restrictions ($\beta, k, h > 0$), show that $\det A$ can never be zero. Explain geometrically why this guarantees a unique equilibrium, and what $\det A = 0$ would have meant.
3. Solve for the equilibrium output Y^* and interest rate i^* using **Cramer's Rule**.
4. **Comparative statics.** An expansionary fiscal policy raises α . Compute the multiplier vector $\frac{\partial z^*}{\partial \alpha}$. Interpret the signs.

Part II: The Non-Linear Generalization

Now consider the more general, non-linear IS-LM model:

$$Y - C(Y) - I(i) - G = 0$$

$$L(Y, i) - \frac{M}{P} = 0$$

Assume standard economic conditions: the marginal propensity to consume $0 < C_Y < 1$, investment is strictly decreasing in the interest rate $I_i < 0$, money demand increases with output $L_Y > 0$, and money demand decreases with the interest rate $L_i < 0$.

5. Totally differentiate the system with respect to the endogenous variables (Y, i) and the exogenous variable G . Set it up in matrix form: $J \begin{pmatrix} dY \\ di \end{pmatrix} = \begin{pmatrix} dG \\ 0 \end{pmatrix}$, where J is the Jacobian matrix.
6. Sign the determinant of the Jacobian ($|J|$). Does a unique local equilibrium exist?
7. **Solving the non-linear system.** You can now *solve* the differentiated system exactly as you solved $Az = b$ in Part I.

- a. Apply **Cramer's Rule** to $J \begin{pmatrix} dY \\ di \end{pmatrix} = \begin{pmatrix} dG \\ 0 \end{pmatrix}$ to obtain $\frac{dY}{dG}$ and $\frac{di}{dG}$ in terms of C_Y, I_i, L_Y, L_i .
- b. Sign both derivatives. Compare them to the signs you found in Part I.
- c. Show that

$$\frac{dY}{dG} < \frac{1}{1 - C_Y},$$

and identify the term in $|J|$ responsible for the gap. What happens to the gap if $I_i = 0$ (investment does not respond to the interest rate)?

- d. Now let the money supply move instead: replace the right-hand side by $\begin{pmatrix} 0 \\ d(M/P) \end{pmatrix}$ and compute $\frac{dY}{d(M/P)}$ and $\frac{di}{d(M/P)}$. Sign them, and say in one sentence how monetary and fiscal expansion differ in their effect on the interest rate.
- e. Take the first row of J and divide it by $(1 - C_Y)$. Compare the result to the matrix A from part 1. What are β , k , and h in terms of the derivatives C_Y, I_i, L_Y, L_i ? Conclude that Part I was the special case of Part II all along.

i Solution

Question 1.

$$A = \begin{pmatrix} 1 & \beta \\ k & -h \end{pmatrix}, \quad z = \begin{pmatrix} Y \\ i \end{pmatrix}, \quad b = \begin{pmatrix} \alpha \\ M/P \end{pmatrix}.$$

Question 2.

$$\det A = (1)(-h) - (\beta)(k) = -(h + \beta k).$$

Since $\beta, k, h > 0$, the determinant is strictly negative ($-(h + \beta k) < 0$), and thus never zero. Geometrically, the IS curve slopes downward in (Y, i) space, while the LM curve slopes upward; they must cross exactly once. If $\det A = 0$, it would imply the two lines are parallel (or collinear), meaning the system has either no equilibrium or infinite equilibria. The economic sign restrictions naturally rule this out.

Question 3.

Using Cramer's Rule for Y^* :

$$Y^* = \frac{\det \begin{pmatrix} \alpha & \beta \\ M/P & -h \end{pmatrix}}{\det A} = \frac{-\alpha h - \beta(M/P)}{-(h + \beta k)} = \frac{\alpha h + \beta(M/P)}{h + \beta k}$$

Using Cramer's Rule for i^* :

$$i^* = \frac{\det \begin{pmatrix} 1 & \alpha \\ k & M/P \end{pmatrix}}{\det A} = \frac{M/P - \alpha k}{-(h + \beta k)} = \frac{\alpha k - M/P}{h + \beta k}$$

Question 4.

The change in the parameter vector is $\frac{\partial b}{\partial \alpha} = (1, 0)^\top$. The inverse matrix is:

$$A^{-1} = \frac{1}{-(h + \beta k)} \begin{pmatrix} -h & -\beta \\ -k & 1 \end{pmatrix}$$

$$\frac{\partial z^*}{\partial \alpha} = A^{-1} \frac{\partial b}{\partial \alpha} = \frac{1}{-(h + \beta k)} \begin{pmatrix} -h \\ -k \end{pmatrix} = \begin{pmatrix} \frac{h}{h + \beta k} \\ \frac{k}{h + \beta k} \end{pmatrix}$$

Both derivatives are strictly positive. Fiscal expansion raises output, but it also pushes the interest rate up: higher output raises money demand, which—with the money supply fixed—can only clear at a higher interest rate. That rise in i is the classic **crowding-out** of private investment.

Question 5.

Taking the total differential of both equations:

$$dY - C_Y dY - I_i di - dG = 0 \implies (1 - C_Y)dY - I_i di = dG$$

$$L_Y dY + L_i di = 0$$

In matrix form:

$$\underbrace{\begin{pmatrix} 1 - C_Y & -I_i \\ L_Y & L_i \end{pmatrix}}_J \begin{pmatrix} dY \\ di \end{pmatrix} = \begin{pmatrix} dG \\ 0 \end{pmatrix}$$

Question 6.

$$|J| = (1 - C_Y)(L_i) - (-I_i)(L_Y) = (1 - C_Y)L_i + I_i L_Y$$

Given $0 < C_Y < 1$, the term $(1 - C_Y)$ is positive. We are given $L_i < 0$, $I_i < 0$, and $L_Y > 0$. Evaluating the signs: $|J| = (+) \times (-) + (-) \times (+) = (-) + (-) < 0$. Since $|J|$ is strictly negative, it is non-zero. By the Implicit Function Theorem, with the Jacobian being invertible at the equilibrium point, guaranteeing a locally unique equilibrium.

Question 7.

(a) With $|J| = (1 - C_Y)L_i + I_i L_Y$, Cramer's Rule replaces the relevant column of J by the right-hand side $(dG, 0)^\top$:

$$dY = \frac{\det \begin{pmatrix} dG & -I_i \\ 0 & L_i \end{pmatrix}}{|J|} = \frac{L_i dG}{|J|}, \quad di = \frac{\det \begin{pmatrix} 1 - C_Y & dG \\ L_Y & 0 \end{pmatrix}}{|J|} = \frac{-L_Y dG}{|J|}.$$

Hence

$$\frac{dY}{dG} = \frac{L_i}{(1 - C_Y)L_i + I_i L_Y}, \quad \frac{di}{dG} = \frac{-L_Y}{(1 - C_Y)L_i + I_i L_Y}.$$

(b) From Question 6, $|J| < 0$. The numerator of dY/dG is $L_i < 0$, so $dY/dG = (-)/(-) > 0$. The numerator of di/dG is $-L_Y < 0$ (since $L_Y > 0$), so $di/dG = (-)/(-) > 0$. Both strictly positive — **exactly the signs of Part I, Question 4**, but now derived without ever assuming C , I or L were linear. The linear algebra never needed the model to be linear; it only needed it to be differentiable, so that the Jacobian could stand in for A locally.

(c) Write the denominator as $|J| = (1 - C_Y)L_i + I_i L_Y$. If the second term were absent we would have

$$\frac{L_i}{(1 - C_Y)L_i} = \frac{1}{1 - C_Y},$$

the textbook Keynesian multiplier. But $I_i L_Y = (-) \times (+) < 0$, so $|J|$ is *more* negative than $(1 - C_Y)L_i$; dividing the same negative numerator L_i by a larger-magnitude negative number gives a *smaller* positive quotient. Hence

$$0 < \frac{dY}{dG} < \frac{1}{1 - C_Y}.$$

The gap is caused entirely by the term $I_i L_Y$: fiscal expansion raises Y , which raises money demand ($L_Y > 0$), which forces i up, which cuts investment ($I_i < 0$). That chain is **crowding out**, and here it is visible as a single product in a determinant.

If $I_i = 0$, that term vanishes, $|J| = (1 - C_Y)L_i$, and $dY/dG = 1/(1 - C_Y)$ *exactly*: the interest rate may still rise, but investment ignores it, so no output is crowded out. This confirms the reading of Part I: crowding out is not “ i goes up,” it is the *shortfall of dY/dG below the multiplier* — the factor $h/(h + \beta k) < 1$ in Question 4.

(d) With right-hand side $(0, d(M/P))^\top$:

$$\frac{dY}{d(M/P)} = \frac{\det \begin{pmatrix} 0 & -I_i \\ 1 & L_i \end{pmatrix}}{|J|} = \frac{I_i}{|J|} = \frac{(-)}{(-)} > 0, \quad \frac{di}{d(M/P)} = \frac{\det \begin{pmatrix} 1 - C_Y & 0 \\ L_Y & 1 \end{pmatrix}}{|J|} = \frac{1 - C_Y}{|J|} = \frac{(+)}{(-)} < 0.$$

Both policies raise output, but they **push the interest rate in opposite directions**: fiscal expansion raises i (crowding out private investment), monetary expansion lowers it (crowding investment *in*). Note that all four multipliers came from the same inverted matrix — one factorisation of J , and the entire policy table is yours.

(e) Dividing the first row of J by $(1 - C_Y) > 0$:

$$J = \begin{pmatrix} 1 - C_Y & -I_i \\ L_Y & L_i \end{pmatrix} \rightarrow \begin{pmatrix} 1 & \frac{-I_i}{1 - C_Y} \\ L_Y & L_i \end{pmatrix} = \begin{pmatrix} 1 & \beta \\ k & -h \end{pmatrix} = A,$$

so

$$\beta = \frac{-I_i}{1 - C_Y} > 0, \quad k = L_Y > 0, \quad h = -L_i > 0,$$

and the sign restrictions $\beta, k, h > 0$ imposed by hand in Part I are *exactly* the economic assumptions $I_i < 0, L_Y > 0, L_i < 0$ of Part II. Since row scaling multiplies the determinant by the same positive factor, $|J| = (1 - C_Y) \det A$, and the two determinants carry the same sign — the same guarantee of a unique (now local) equilibrium. Part I was Part II linearised.

Eigenvalues, Diagonalisation, and Dynamic Stability

Eigenvalues, Diagonalisation

A nonzero v with $Av = \lambda v$ is an **eigenvector** with **eigenvalue** λ ; the λ are the roots of $\det(A - \lambda I) = 0$. If A has n independent eigenvectors it is **diagonalisable**, $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, and then $A^k = PD^kP^{-1}$. Also $\det A = \prod_i \lambda_i$ and $\text{tr } A = \sum_i \lambda_i$.

Stability of a Linear System

For $x_{t+1} = Ax_t$ (**discrete time**), the origin is stable iff every eigenvalue satisfies $|\lambda_i| < 1$. For $\dot{x} = Ax$ (**continuous time**), it is stable iff every eigenvalue satisfies $\text{Re}(\lambda_i) < 0$. Mixed cases (some roots stable, some not) give a **saddle**.

Exercise 9 : Diagonalisation and matrix powers

Let $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

1. Find the eigenvalues of A , then an eigenvector for each.
2. Write $A = PDP^{-1}$ explicitly (give P , D , and P^{-1}), and verify $P^{-1}AP = D$.
3. Use the eigenvalues to read off $\det A$ and $\text{tr } A$, and write the general formula for A^k . Explain the computational payoff of diagonalisation for large k .
4. Explain in one or two sentences why any 2×2 matrix with two **distinct** eigenvalues is automatically diagonalisable.

Solution

Question 1.

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2).$$

Eigenvalues $\lambda_1 = 5$, $\lambda_2 = 2$.

For $\lambda_1 = 5$: $(A - 5I) = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$, so $-v_1 + v_2 = 0$, giving $v_1 = (1, 1)$.

For $\lambda_2 = 2$: $(A - 2I) = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$, so $2v_1 + v_2 = 0$, giving $v_2 = (1, -2)$.

Question 2.

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}, \quad \det P = -3, \quad P^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

Verification: $AP = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 5 & 2 \\ 5 & -4 \end{pmatrix}$, and

$$P^{-1}(AP) = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 5 & 2 \\ 5 & -4 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 15 & 0 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix} = D. \checkmark$$

Question 3.

$\det A = \lambda_1 \lambda_2 = 10$ (check: $4 \cdot 3 - 1 \cdot 2 = 10$) and $\text{tr } A = \lambda_1 + \lambda_2 = 7$ (check: $4 + 3 = 7$). The power formula is

$$A^k = PD^kP^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 5^k & 0 \\ 0 & 2^k \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

For large k , computing A^k directly costs $k - 1$ matrix multiplications; via PD^kP^{-1} it costs raising two scalars to the k -th power plus a fixed three matrix multiplications — essentially free by comparison.

Question 4.

Eigenvectors belonging to distinct eigenvalues are always linearly independent. Two distinct eigenvalues therefore yield two independent eigenvectors, which is exactly $n = 2$ of them — enough to fill the columns of an invertible P , so A is diagonalisable.

Exercise 10 : A saddle-path economy and the Blanchard–Kahn logic

This exercise is entirely self-contained: you do not need any macroeconomics to do it.. We build up the economic story in small steps before asking you to touch a single matrix.

Setting the scene. Imagine an economy described at each date $t = 0, 1, 2, \dots$ by two numbers stacked in a vector $x_t = (x_{1,t}, x_{2,t})^\top$:

- $x_{1,t}$ is the **capital stock**. Capital is a physical quantity of machines and buildings: it is whatever was inherited from yesterday's investment decisions, and it cannot jump. In particular, $x_{1,0} = k_0$ is simply *given* to you as data — a number fixed by history, not something anyone chooses at date 0. Economists call this a **predetermined** variable.
- $x_{2,t}$ is some **price** (think of it as an asset price, or an exchange rate). Unlike capital, a price can move instantly: nothing physical stops it from taking any value at date 0. Its initial value $x_{2,0}$ is a **free choice** — economists call it a **jump variable**.

The economy evolves according to the linear law of motion $x_{t+1} = Ax_t$ with

$$A = \begin{pmatrix} 1.25 & 0.75 \\ 0.75 & 1.25 \end{pmatrix}.$$

Once $x_0 = (x_{1,0}, x_{2,0})$ is fixed, every future x_t is completely determined by repeatedly applying A . The economic question is: **which value of the free jump variable $x_{2,0}$** should we expect the market to pick? We will see that linear algebra answers this question, and that (perhaps surprisingly) *almost every* choice of $x_{2,0}$ leads to nonsense.

To simplify the quantities, both entries are normalized as deviations from a long-run steady state, so $x_{i,t} = 0$ means “at its steady-state value” and negative entries are perfectly sensible.

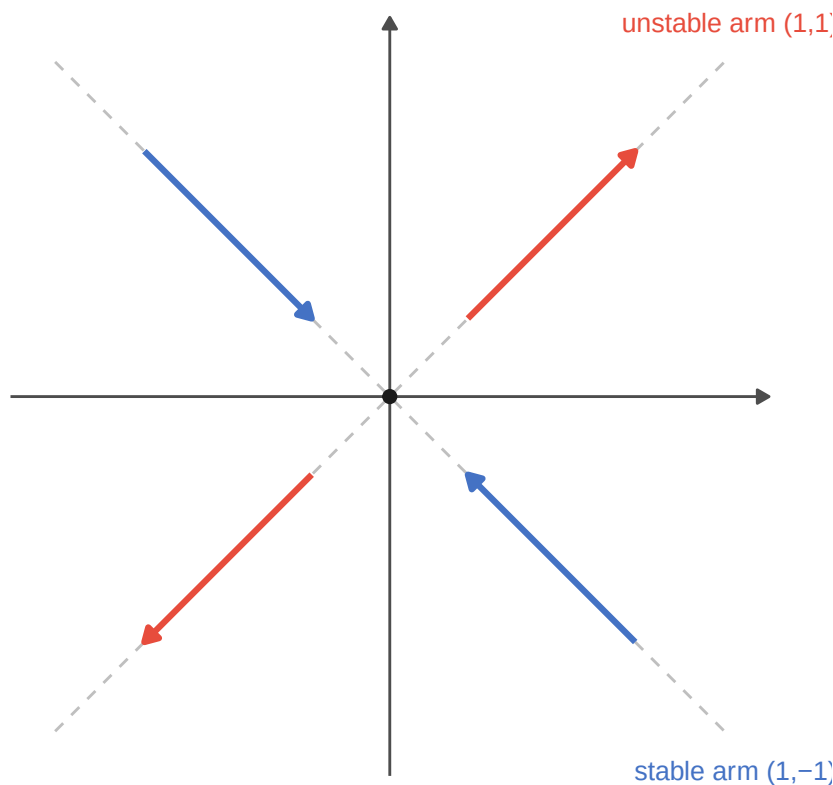
1. Before touching A , consider the trivial one-dimensional example $y_{t+1} = 1.5 y_t$ with $y_0 = 1$. Compute y_1, y_2, y_3, y_{10} . What happens to y_t as $t \rightarrow \infty$? Now do the same for $z_{t+1} = 0.5 z_t$ with $z_0 = 1$. Contrast the two behaviours in one sentence.
2. **Diagonalise A .** Find the eigenvalues λ_1, λ_2 of A and an eigenvector for each. You should get one eigenvalue bigger than 1 in absolute value and one smaller.
3. **Classify the two directions.** In discrete time, a component of the dynamics that moves along an eigenvector with $|\lambda| > 1$ grows without bound; a component along an eigenvector with $|\lambda| < 1$ shrinks to zero. Using this, label which of your two eigenvectors spans the **stable arm** (deviations die out) and which spans the **unstable arm** (deviations explode). Is the origin of this system stable, unstable, or a saddle?

4. **Write down the general path.** Any initial condition x_0 can be written as a combination of the two eigenvectors, $x_0 = c_1 v_1 + c_2 v_2$. Show that this implies

$$x_t = c_1 \lambda_1^t v_1 + c_2 \lambda_2^t v_2.$$

(Hint: apply A to x_0 once, using $Av_i = \lambda_i v_i$, and spot the pattern; or simply note $x_t = A^t x_0$ and diagonalise as in Exercise 9.)

5. **Why almost every starting point is economically absurd.** Suppose $c_1 \neq 0$ (the coefficient on the *unstable* eigenvector). What happens to $\|x_t\|$ as $t \rightarrow \infty$? Explain briefly why no sensible equilibrium can look like this. Conclude that a bounded, economically sensible path requires $c_1 = 0$, i.e. x_0 must lie **exactly on the stable arm**.
6. **Solving for the jump variable.** The capital stock $x_{1,0} = k_0$ is fixed by history — you do not get to choose it. But being on the stable arm pins down a *specific* relationship between $x_{1,0}$ and $x_{2,0}$. Use the stable eigenvector you found in Question 3 to solve for the unique $x_{2,0}$ (as a function of k_0) that places x_0 on the stable arm. This is the value the jump variable “jumps to” at date 0.
7. **Putting the pieces together.** In one or two sentences, explain why the argument above delivers a **unique** economically sensible equilibrium: one predetermined variable, one unstable eigenvalue, and a jump variable that has exactly one degree of freedom to spend. (This equality between “number of jump variables” and “number of unstable roots” is known as the **Blanchard–Kahn condition** for an economy to follow a stable course)



i Solution

Question 1.

$y_0 = 1$, $y_1 = 1.5$, $y_2 = 2.25$, $y_3 = 3.375$, and $y_{10} = 1.5^{10} \approx 57.7$. As $t \rightarrow \infty$, $y_t = 1.5^t \rightarrow +\infty$: the sequence **explodes**, because we keep multiplying by a number bigger than 1. For z : $z_0 = 1$, $z_1 = 0.5$, $z_2 = 0.25$, $z_3 = 0.125$, and $z_t = 0.5^t \rightarrow 0$: the sequence **shrinks**

to zero, because we keep multiplying by a number smaller than 1 (in absolute value). The contrast: multiplying repeatedly by a number with $|\lambda| > 1$ blows a sequence up; multiplying repeatedly by a number with $|\lambda| < 1$ kills it. This is *all* that will happen inside each eigen-direction of A — the eigenvalue plays exactly the role that 1.5 or 0.5 played here.

Question 2.

$\text{tr } A = 1.25 + 1.25 = 2.5$, $\det A = 1.25^2 - 0.75^2 = 1.5625 - 0.5625 = 1$. The characteristic polynomial $\lambda^2 - 2.5\lambda + 1 = 0$ factors as $(\lambda - 2)(\lambda - 0.5)$, so $\lambda_1 = 2$, $\lambda_2 = 0.5$. As promised: one root above 1, one below.

For $\lambda_1 = 2$: $(A - 2I) = \begin{pmatrix} -0.75 & 0.75 \\ 0.75 & -0.75 \end{pmatrix}$. The first row gives $-0.75 v_1 + 0.75 v_2 = 0$, i.e. $v_1 = v_2$, so an eigenvector is $(1, 1)$.

For $\lambda_2 = 0.5$: $(A - 0.5I) = \begin{pmatrix} 0.75 & 0.75 \\ 0.75 & 0.75 \end{pmatrix}$. The first row gives $0.75 v_1 + 0.75 v_2 = 0$, i.e. $v_1 = -v_2$, so an eigenvector is $(1, -1)$.

Question 3.

Exactly as in the warm-up: the component of x_t that sits along $(1, 1)$ gets multiplied by $\lambda_1 = 2$ every period, so it doubles each time — it **explodes**. The component along $(1, -1)$ gets multiplied by $\lambda_2 = 0.5$ every period, so it halves each time — it **shrinks to zero**. So $\text{span}\{(1, -1)\}$ is a **stable arm** and $\text{span}\{(1, 1)\}$ is the **unstable arm**. Since one direction attracts and the other repels, the origin is neither stable nor unstable outright: it is a **saddle point** $\rightarrow$ the price must come from the stable arm.

Question 4.

Write $x_0 = c_1(1, 1) + c_2(1, -1)$. Applying A once, and using $Av_i = \lambda_i v_i$ for each eigenvector separately:

$$x_1 = Ax_0 = c_1 A(1, 1) + c_2 A(1, -1) = c_1 \lambda_1(1, 1) + c_2 \lambda_2(1, -1) = c_1 2(1, 1) + c_2 0.5(1, -1).$$

Applying A repeatedly (or, equivalently, diagonalising $A = PDP^{-1}$ as in Exercise 9 and using $A^t = PD^tP^{-1}$) gives, after t steps,

$$x_t = c_1 2^t(1, 1) + c_2 (0.5)^t(1, -1).$$

Each eigen-direction simply gets multiplied by its own eigenvalue, t times in a row — nothing more than the warm-up repeated twice, once per direction.

Question 5.

If $c_1 \neq 0$, the term $c_1 2^t(1, 1)$ grows without bound as $t \rightarrow \infty$ (exactly like y_t in Question 1), while the other term shrinks to zero. So $\|x_t\| \rightarrow \infty$: both the capital stock and the price would grow (or swing) towards infinity forever. This is not a sensible description of an economy — capital cannot literally become infinite in finite time, and no rational household would hold an asset whose price is expected to diverge without bound relative to everything else. We rule such paths out.

The only way to avoid this is $c_1 = 0$. Then $x_0 = c_2(1, -1)$, which means x_0 must lie exactly on the stable arm — a very special, one-dimensional set of starting points inside the whole two-dimensional plane of *conceivable* starting points.

Question 6.

x_0 on the stable arm means $x_0 = c_2(1, -1)$ for some c_2 , i.e. $x_{1,0} = c_2$ and $x_{2,0} = -c_2$. Since $x_{1,0} = k_0$ is given, this forces $c_2 = k_0$ and hence

$$x_{2,0} = -k_0.$$

That is the one and only value the jump variable can take at date 0 if the economy is to avoid exploding: given whatever capital stock history has handed us, the price must “jump” immediately to $-k_0$.

Question 7.

Capital $x_{1,0} = k_0$ is fixed — no degree of freedom there. The jump variable $x_{2,0}$ *does* have one degree of freedom (any real number, in principle), and there is exactly one unstable eigenvalue. Question 6 showed that this single degree of freedom is exactly enough to cancel the coefficient c_1 on the unstable direction, and no more: the jump variable is not free to do anything else, because any other choice reintroduces the explosive component from Question 5. This “one jump variable per unstable root” matching is precisely the **Blanchard–Kahn condition**: it is what guarantees existence *and* uniqueness of a bounded equilibrium path in linearised macro models. Too few jump variables relative to unstable roots and no bounded solution exists at all; too many, and there are infinitely many.

Exercise 11 : A Markov chain and the long-run unemployment rate

Workers move each period between **Employed** (E) and **Unemployed** (U). An employed worker stays employed with probability 0.9 and loses the job with probability 0.1; an unemployed worker finds a job with probability 0.6 and stays unemployed with probability 0.4.

Setting up the bookkeeping. At date t , let π_t^E be the *share* of the whole workforce that is currently employed, and π_t^U the share currently unemployed, so $\pi_t^E + \pi_t^U = 1$. Stack them into $\pi_t = (\pi_t^E, \pi_t^U)^\top$. Each period, a slice of the employed pool becomes unemployed and a slice of the unemployed pool becomes employed, so tomorrow’s shares are a weighted blend of today’s:

$$\pi_{t+1}^E = 0.9 \pi_t^E + 0.6 \pi_t^U, \quad \pi_{t+1}^U = 0.1 \pi_t^E + 0.4 \pi_t^U.$$

In matrix form this is $\pi_{t+1} = P\pi_t$ with

$$P = \begin{pmatrix} 0.9 & 0.6 \\ 0.1 & 0.4 \end{pmatrix}.$$

We want to know: if we let this process run forever, does the unemployment rate settle down to some fixed number, and if so, which number — regardless of where we started?

1. **Get your hands dirty first.** Suppose the whole workforce starts out employed: $\pi_0 = (1, 0)$. Compute $\pi_1 = P\pi_0$, then $\pi_2 = P\pi_1$, then $\pi_3 = P\pi_2$ by direct matrix multiplication (no eigenvalues yet). What do you notice about $\pi_1^U, \pi_2^U, \pi_3^U$ — are they moving towards some number? (*You don’t need to guess the exact limit yet — just observe the trend.*)
2. **Read the matrix.** Confirm that each column of P sums to 1 (this property is called **column-stochastic**), and explain in words what column 1 and column 2 each represent in terms of where a worker goes next period.
3. **Why $\lambda = 1$ is special here.** A **stationary distribution** π^* is a distribution that reproduces itself: $P\pi^* = \pi^*$. Recognise that this is exactly the eigenvector equation $P\pi^* = \lambda\pi^*$ with $\lambda = 1$. So finding the long-run outcome (if it exists) reduces to finding an eigenvector of P with eigenvalue 1.
4. $\lambda = 1$ is guaranteed to exist. Before computing anything, show that $\lambda = 1$ is *always* an eigenvalue of a column-stochastic matrix like P . (*Hint: compute $\mathbf{1}^\top P$ where $\mathbf{1} = (1, 1)^\top$, using the column-sums-to-1 property from Question 2. What does this tell you about $\mathbf{1}$ as a left eigenvector of P ? You may use, without proof, the fact that P and $P^\top$ share the same eigenvalues.*)
5. **Solve for the stationary distribution.** Solve $(P - I)\pi = 0$ for π , then normalise so that $\pi^E + \pi^U = 1$ (a distribution’s shares must sum to 1). Call the result π^* . Compare $\pi^{*,U}$ to the trend you spotted numerically in Question 1.

6. **The second eigenvalue.** Find the other eigenvalue λ_2 of P (using $\text{tr } P$ and $\det P$, as in Exercise 9). Confirm $|\lambda_2| < 1$.
7. **Why any starting point converges — step by step.** Any initial distribution π_0 can be written as a combination of the two eigenvectors, $\pi_0 = a\pi^* + bv_2$, where v_2 is an eigenvector for λ_2 .
- Using $P\pi^* = \pi^*$ and $Pv_2 = \lambda_2 v_2$, write down $\pi_t = P^t\pi_0$ in terms of a , b , π^* , v_2 , λ_2 , and t .
 - What happens to the v_2 term as $t \rightarrow \infty$, given $|\lambda_2| < 1$? (Compare to the warm-up reasoning from Exercise 10, Question 1.)
 - It remains to pin down a . Show that the entries of v_2 must sum to 0. (Hint: apply $\mathbf{1}^\top$ to both sides of $Pv_2 = \lambda_2 v_2$, use $\mathbf{1}^\top P = \mathbf{1}^\top$ from Question 4, and solve for $\mathbf{1}^\top v_2$ given that $\lambda_2 \neq 1$.)
 - Every π_t is a distribution, so its two entries must sum to 1 for **every** t , including $t = 0$. Using part (c) and $\pi^{*,E} + \pi^{*,U} = 1$, show this forces $a = 1$.
 - Conclude: what is $\lim_{t \rightarrow \infty} \pi_t$? Does it depend on the starting point π_0 ?
8. **Speed of convergence.** Explain, in one sentence, why a *smaller* $|\lambda_2|$ means the economy forgets its starting unemployment rate *faster*. Roughly how many periods does it take for the gap $\|\pi_t - \pi^*\|$ to shrink by a factor of 10, given $\lambda_2 = 0.3$? (A rough order of magnitude is fine — you may use $\log_{10}$.)

i Solution

Question 1.

$$\begin{aligned}\pi_1 &= P\pi_0 = \begin{pmatrix} 0.9 & 0.6 \\ 0.1 & 0.4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}. \\ \pi_2 &= P\pi_1 = \begin{pmatrix} 0.9(0.9) + 0.6(0.1) \\ 0.1(0.9) + 0.4(0.1) \end{pmatrix} = \begin{pmatrix} 0.87 \\ 0.13 \end{pmatrix}. \\ \pi_3 &= P\pi_2 = \begin{pmatrix} 0.9(0.87) + 0.6(0.13) \\ 0.1(0.87) + 0.4(0.13) \end{pmatrix} = \begin{pmatrix} 0.861 \\ 0.139 \end{pmatrix}.\end{aligned}$$

So $\pi_1^U = 0.10$, $\pi_2^U = 0.13$, $\pi_3^U = 0.139$: the unemployment share is climbing from 0 but by smaller and smaller steps each period (+0.10, then +0.03, then +0.009), as if it is homing in on some fixed number a bit above 0.13. That is exactly the long-run rate we will compute exactly in Question 5.

Question 2.

Column 1 sums to $0.9 + 0.1 = 1$ and column 2 sums to $0.6 + 0.4 = 1$, so P is column-stochastic. Column 1 lists where a **currently employed** worker goes next period (stays E with probability 0.9, becomes U with probability 0.1); column 2 lists where a **currently unemployed** worker goes next period (becomes E with probability 0.6, stays U with probability 0.4). Each column is a full set of destination probabilities for one starting state, which is why it must sum to 1: the worker has to go *somewhere*.

Question 3.

$P\pi^* = \pi^*$ says applying P to π^* gives back π^* unchanged — multiplying by 1. That is literally the definition of an eigenvector with eigenvalue $\lambda = 1$: $P\pi^* = 1 \cdot \pi^*$. So “find the distribution that reproduces itself” and “find the eigenvector of P for $\lambda = 1$ ” are the same task.

Question 4.

Column-stochasticity (Question 2) says every column of P sums to 1; in vector form this is exactly $\mathbf{1}^\top P = \mathbf{1}^\top$ (multiplying $\mathbf{1}^\top$ against P sums each column). This equation says $\mathbf{1}$ is a **left** eigenvector of P with eigenvalue 1: $\mathbf{1}^\top P = 1 \cdot \mathbf{1}^\top$. Since P and $P^\top$ always have the same eigenvalues, 1 is also an eigenvalue of P itself — with some (right) eigenvector, which is exactly the stationary distribution we are looking for. So we didn't get lucky: **every** column-stochastic matrix has $\lambda = 1$ as an eigenvalue, guaranteed by the column-sum-to-1 property alone.

Question 5.

Solve $(P - I)\pi = 0$:

$$P - I = \begin{pmatrix} -0.1 & 0.6 \\ 0.1 & -0.6 \end{pmatrix} \implies -0.1\pi^E + 0.6\pi^U = 0 \implies \pi^E = 6\pi^U.$$

Normalising $\pi^E + \pi^U = 1$: substituting gives $6\pi^U + \pi^U = 1$, so $7\pi^U = 1$, hence

$$\pi^* = \left(\frac{6}{7}, \frac{1}{7}\right) \approx (0.857, 0.143).$$

The long-run unemployment rate is $\pi^{*,U} = 1/7 \approx 14.3\%$ — consistent with the numerical trend from Question 1, which was climbing towards something just above 0.139.

Question 6.

$\text{tr } P = 0.9 + 0.4 = 1.3$ and $\det P = (0.9)(0.4) - (0.6)(0.1) = 0.36 - 0.06 = 0.30$. The characteristic polynomial is $\lambda^2 - 1.3\lambda + 0.3 = 0$, which factors as $(\lambda - 1)(\lambda - 0.3) = 0$ (we already know $\lambda = 1$ is a root from Question 4). So $\lambda_2 = 0.3$, and indeed $|\lambda_2| = 0.3 < 1$.

Question 7.

(a) Since $P\pi^* = \pi^*$ and $Pv_2 = \lambda_2 v_2$, applying P to $\pi_0 = a\pi^* + bv_2$ gives $P\pi_0 = a\pi^* + b\lambda_2 v_2$; repeating this t times (or diagonalising as in Exercise 9) gives

$$\pi_t = P^t \pi_0 = a \cdot 1^t \pi^* + b(\lambda_2)^t v_2 = a\pi^* + b(0.3)^t v_2.$$

(b) Since $|\lambda_2| = 0.3 < 1$, we have $(0.3)^t \rightarrow 0$ as $t \rightarrow \infty$ — exactly the shrinking behaviour from the $z_t = 0.5^t$ warm-up in Exercise 10, Question 1, just with a different base. So the whole term $b(0.3)^t v_2$ vanishes, no matter what b is.

(c) Apply $\mathbf{1}^\top$ to both sides of $Pv_2 = \lambda_2 v_2$:

$$\mathbf{1}^\top Pv_2 = \lambda_2 \mathbf{1}^\top v_2.$$

But $\mathbf{1}^\top P = \mathbf{1}^\top$ from Question 4, so the left side is just $\mathbf{1}^\top v_2$. Hence

$$\mathbf{1}^\top v_2 = \lambda_2 \mathbf{1}^\top v_2 \implies (1 - \lambda_2) \mathbf{1}^\top v_2 = 0.$$

Since $\lambda_2 = 0.3 \neq 1$, the factor $(1 - \lambda_2)$ is nonzero, which forces $\mathbf{1}^\top v_2 = 0$ — i.e. the two entries of v_2 sum to 0.

(d) Every π_t is a distribution of shares, so its entries sum to 1 for every t — in particular at $t = 0$: $\mathbf{1}^\top \pi_0 = 1$. Apply $\mathbf{1}^\top$ to $\pi_0 = a\pi^* + bv_2$:

$$\mathbf{1}^\top \pi_0 = a \mathbf{1}^\top \pi^* + b \mathbf{1}^\top v_2 = a \cdot 1 + b \cdot 0 = a,$$

using $\pi^{*,E} + \pi^{*,U} = 1$ from Question 5 and $\mathbf{1}^\top v_2 = 0$ from part (c). Since $\mathbf{1}^\top \pi_0 = 1$, this gives $a = 1$.

(e) Combining (a), (b), (d):

$$\pi_t = 1 \cdot \pi^* + b(0.3)^t v_2 \xrightarrow{t \rightarrow \infty} \pi^*.$$

The limit is π^* regardless of b — and b is precisely what encoded the starting point π_0 . So **every** initial distribution converges to the same $\pi^* = (6/7, 1/7)$: the long-run unemployment rate does not depend on where the labour market started.

Question 8.

A smaller $|\lambda_2|$ means the factor $(\lambda_2)^t$ shrinks to zero faster as t grows, so the “memory” of the initial distribution $b(\lambda_2)^t v_2$ fades more quickly — the chain forgets its starting point sooner. To shrink by a factor of 10 we need $(0.3)^t \approx 0.1$, i.e. $t \log_{10}(0.3) \approx \log_{10}(0.1) = -1$,

so $t \approx \frac{-1}{\log_{10}(0.3)} = \frac{-1}{-0.523} \approx 1.9$ — roughly **two periods** for the gap to shrink tenfold.

Convergence here is fast precisely because $|\lambda_2| = 0.3$ is well below 1.

OLS as a Projection

OLS and the Hat Matrix

For $Y = X\beta + \varepsilon$ with $X \in \mathbb{M}_{N \times K}$ of full column rank, the OLS estimator minimises $\|Y - Xb\|^2$ and equals

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y, \quad \hat{Y} = P_X Y, \quad \hat{\varepsilon} = M_X Y,$$

with $P_X = X(X^\top X)^{-1} X^\top$ and $M_X = I - P_X$. Residuals are orthogonal to every regressor: $X^\top \hat{\varepsilon} = 0$.

Frisch–Waugh–Lovell

In the partitioned model $Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$, with $M_1 = I - P_{X_1}$,

$$\hat{\beta}_2 = (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 Y.$$

Equivalently: partial X_1 out of both Y and X_2 , then regress the residuals — the coefficient on X_2 is identical to its value in the full regression.

Exercise 12 : Deriving OLS and the orthogonal residual

i Motivation

Suppose you’ve collected data on three students. For each one you know how many **hours they studied** and what **exam score** they got:

student	hours studied	exam score
A	1	1
B	2	3
C	3	4

You suspect score goes up roughly linearly with hours, so you want to draw the “best” straight line $\text{score} \approx \beta_1 + \beta_2 \cdot (\text{hours})$ through these three points. Here β_1 is the intercept (predicted score for zero hours) and β_2 is the slope (extra score per extra hour of study). The catch: no straight line passes through all three points exactly. Whatever line you pick, it misses each point by some amount: call those three misses the **prediction errors** ε .

OLS (“ordinary least squares”) is about following a simple rule: *pick the line that makes the total of the squared misses as small as possible*.

In this exercise, we will make clear that the best line is actually an orthogonal projection.

Collect the scores in a vector $Y = (1, 3, 4)$ and stack the inputs in a matrix X whose first column is all 1’s (that column represents the fact that individuals would have had some test score, even in the absence of study time) and whose second column is the hours $(1, 2, 3)$:

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}.$$

1. **Find the best coefficients.** We want the b minimising the total squared error $L(b) = \|Y - Xb\|^2$.

- a. Expand $L(b) = (Y - Xb)^\top(Y - Xb)$ into three terms in b .
 - b. To minimise, differentiate with respect to b , set the gradient to zero, and solve to obtain the **normal equations** $X^\top Xb = X^\top Y$ and $\hat{\beta} = (X^\top X)^{-1}X^\top Y$. *Hint: you can either do the derivatives directly with the matrices if you are comfortable, or expand into 2 equations and derivate each equation separately.*
 - c. Solving needs $X^\top X$ to be invertible. Look at the two columns of X and explain why they're independent. Explain why linear independence of columns of X is sufficient to have invertibility of $X^\top X$. *Hint: think about what linear independence means for the equations $X^\top Xb = 0$ and $Xb = 0$.*
 - d. Suppose a careless research assistant adds a third column recording study time **in minutes**, so X becomes 3×3 with columns $(1, 1, 1)$, $(1, 2, 3)$ and $(60, 120, 180)$. Find a nonzero b with $Xb = 0$. What does exercise 5, Question 4 then say about how many b solve the normal equations?
 - e. Even so, the *predictions* are fine. Explain why $\text{col}(X)$ is the same set with or without the minutes column, and conclude that $\hat{Y}$ is unchanged. Which of $\hat{\beta}$ and $\hat{Y}$ is the fragile one, and why?
 - f. A subtler failure, and the one you will actually meet: to a regression with an intercept, add a dummy D_M equal to 1 for male students and 0 otherwise, *and* a dummy D_F equal to 1 for female students and 0 otherwise. Show the columns of X are dependent. Conclude.
2. Can you explain why the orthogonal projection solves the minimization problem?
 3. **Crunch the numbers.**
 - a. Compute $X^\top X$ and $X^\top Y$.
 - b. Invert $X^\top X$ and get $\hat{\beta} = (\beta_1, \beta_2)$. State, in words, what your intercept and slope mean in this exercise.
 - c. Compute the fitted scores $\hat{Y} = X\hat{\beta}$ and the residuals $\hat{\varepsilon} = Y - \hat{Y}$.
 - d. Check that $X^\top \hat{\varepsilon} = 0$ (both entries). The second entry vanishing is exactly “errors uncorrelated with hours studied.”

i Solution

Question 1.

(a) Multiplying out, and noting $b^\top X^\top Y$ is a single number equal to its transpose $Y^\top Xb$ (so the two cross-terms merge):

$$L(b) = (Y - Xb)^\top(Y - Xb) = Y^\top Y - 2b^\top X^\top Y + b^\top X^\top Xb.$$

(b) Differentiate (rule $\nabla_b(b^\top c) = c$ on the middle term, $\nabla_b(b^\top Mb) = 2Mb$ on the last):

$$\nabla_b L = -2X^\top Y + 2X^\top Xb = 0 \implies X^\top Xb = X^\top Y \implies \hat{\beta} = (X^\top X)^{-1}X^\top Y.$$

(c) The columns are $(1, 1, 1)$ and $(1, 2, 3)$. Neither is a scalar multiple of the other — the first is constant while the second increases — so they're linearly independent, X has full column rank, and $X^\top X$ is positive definite, hence invertible.

(d) Minutes are hours times 60, so the third column is 60 times the second. With $b = (0, 60, -1)^\top$:

$$Xb = 60(1, 2, 3) - (60, 120, 180) = 0.$$

So $\ker(X) \neq \{0\}$, and by (ii) $\ker(X^\top X) \neq \{0\}$ as well. By Exercise 5, Question 4, a system whose matrix has a nontrivial kernel has either no solution or infinitely many; the normal equations always have at least one, so here it is **infinitely many**. If $(\beta_1, \beta_2, \beta_3)$ fits, so does $(\beta_1, \beta_2 + 60t, \beta_3 - t)$ for any real t . There is not “the” coefficient on hours any more. (e) $\text{col}(X)$ is the set of everything you can build from the columns. Since the minutes column is $60\times$ the hours column, anything built with minutes could already be built with hours by multiplying the weight by 60. The extra column adds no new direction, so the column space is the same plane as before.

The projection $\hat{Y}$ depends only on that plane: it is the point of the plane closest to Y . So $\hat{Y}$, the residuals and the fit are all unchanged. What breaks is our ability to *name* $\hat{Y}$ (i.e. find a unique combination to express it): β is the list of weights expressing $\hat{Y}$ in terms of the columns, and weights are unique only when the columns are independent. So $\hat{Y}$ is robust, fixed by geometry; $\hat{\beta}$ is fragile.

(f) Every student is either male or female, so row by row $D_M + D_F = 1$: the two dummy columns add to the all-ones column. With $b = (1, -1, -1)^\top$ on the ordering (intercept, D_M, D_F),

$$Xb = \mathbf{1} - D_M - D_F = 0,$$

so the three columns are dependent. Note that *no two* of them are proportional. Independence is a condition on all the columns **jointly**. The fix is to drop one: keeping the intercept and D_M , the intercept is now the female baseline and β_{D_M} is interpreted as the male–female gap.

Question 2.

Every candidate prediction Xb is a vector in the column space $\text{col}(X)$ — the set of all combinations you can build out of the regressors (here, the ones column and the hours column). Minimising $L(b) = \|Y - Xb\|^2$ is therefore a purely geometric problem: *among all points of $\text{col}(X)$, find the one closest to the data vector Y .*

The closest point of a subspace to an outside point is the foot of the perpendicular dropped onto that subspace — the **orthogonal projection**. Any other point of $\text{col}(X)$ is strictly farther away: if the residual $Y - Xb$ still had a component lying inside $\text{col}(X)$, then sliding Xb a little along that component would shorten the distance, so b could not have been the minimiser. Hence the minimiser $\hat{Y} = X\hat{\beta}$ is exactly the projection $P_X Y$, and its residual $\hat{\varepsilon} = Y - \hat{Y}$ is orthogonal to every column of X :

$$X^\top \hat{\varepsilon} = 0.$$

That orthogonality is not an extra assumption bolted on afterwards — it *is* the geometric meaning of “closest,” and it is what we verify by direct computation in Question 3(d).

Question 3.

(a)

$$X^\top X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad X^\top Y = \begin{pmatrix} 1 + 3 + 4 \\ 1 \cdot 1 + 2 \cdot 3 + 3 \cdot 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}.$$

(b) $\det(X^\top X) = 42 - 36 = 6$, so $(X^\top X)^{-1} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$ and

$$\hat{\beta} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix} \begin{pmatrix} 8 \\ 19 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 112 - 114 \\ -48 + 57 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} -2 \\ 9 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} \\ \frac{3}{2} \end{pmatrix}.$$

So the best line is score $\approx -\frac{1}{3} + \frac{3}{2}$ (hours): the slope $\frac{3}{2}$ says each extra hour of study buys about 1.5 more points, and the intercept $-\frac{1}{3}$ is the (not physically meaningful, slightly

negative) predicted score at zero hours — a reminder that intercepts are often just where the fitted line happens to cross, not a real prediction.

(c) Fitted scores $\hat{Y} = X\hat{\beta}$ at hours 1, 2, 3: $-\frac{1}{3} + \frac{3}{2} = \frac{7}{6}$, $-\frac{1}{3} + 3 = \frac{16}{6}$, $-\frac{1}{3} + \frac{9}{2} = \frac{25}{6}$. Hence

$$\hat{Y} = \left(\frac{7}{6}, \frac{16}{6}, \frac{25}{6}\right), \quad \hat{\varepsilon} = Y - \hat{Y} = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right).$$

(Student B is under-predicted by $\frac{1}{3}$; A and C are each over-predicted by $\frac{1}{6}$.)

(d) Ones column: $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = \frac{-1+2-1}{6} = 0$. Hours column: $1(-\frac{1}{6}) + 2(\frac{1}{3}) + 3(-\frac{1}{6}) = \frac{-1+4-3}{6} = 0$. Both entries of $X^\top \hat{\varepsilon}$ vanish. ✓ The errors sum to zero *and* are uncorrelated with hours — exactly what “best line” forced on us in Q2.